Labeling Efficiency-QC

In the experiment, 1μg RNA is used for labeling. The specific activity (pmol dyes per μg cRNA) of the labeled RNA can be obtained by the following calculation:

$$\text{Specific Activity} = \frac{\text{(pmol per μl dye)}}{\text{(μg per μl cRNA)}}$$

Sample ID	Dye Name	Dye pmol/μl	cRNA Concentration (μg/μl)	Specific Activity*(pmol Dye/μg cRNA)	Volume (μl)	Total Amount (μg)
Mock-1 sup	Cy3	7.65	0.69	11.05	10.00	6.92
Mock-2 sup	Cy3	7.19	0.65	10.98	10.00	6.55
Mock-3 sup	Cy3	9.93	0.73	13.53	10.00	7.34
PR8-1 sup	Cy3	9.24	0.70	13.15	10.00	7.02
PR8-2 sup	Cy3	10.04	0.79	12.66	10.00	7.93
PR8-3 sup	Cy3	10.88	0.78	13.91	10.00	7.82
Mock-1 ip	Cy5	3.47	0.33	10.40	10.00	3.34
Mock-2 ip	Cy5	3.43	0.33	10.24	10.00	3.35
Mock-3 ip	Cy5	3.92	0.35	11.18	10.00	3.51
PR8-1 ip	Cy5	4.22	0.38	11.07	10.00	3.81
PR8-2 ip	Cy5	3.05	0.29	10.50	10.00	2.90
PR8-3 ip	Cy5	3.06	0.32	9.70	10.00	3.16

*For two-color, if the yield is <825 ng and the specific activity is <8.0 pmol Cy3 or Cy5 per μg cRNA do not proceed to the hybridization step. Repeat cRNA preparation.

*For one-color, if the yield is <1.65μg and the specific activity is <9.0 pmol Cy3 or Cy5 per μg cRNA do not proceed to the hybridization step. Repeat cRNA preparation.